

Dear Editors:

We would like to submit the enclosed manuscript entitled “The Vanishing First Rung? Generative AI, Organisational Learning and Youth Entry Employment in a Socio-Technical System Dynamics Model” for consideration in the Special Issue “Systems Approaches to Generative AI: Workforce Development, Organisational Learning, and Economic Transformation” of *Systems*. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by the author.

Generative AI is diffusing fastest through exactly the codified, routine-cognitive tasks that have traditionally formed the entry port of a professional career, and the first high-frequency evidence shows employment among workers aged 22–25 in exposed occupations falling by 13–16% while employment of their experienced colleagues is unchanged. The manuscript argues that this is not a technological verdict but a system outcome, and it builds and validates the socio-technical system dynamics model needed to establish that claim. The entry port is simultaneously a factor of production and the intake stage of the firm's own capability pipeline, so its contraction feeds back — with a delay of the order of the time to competence — on organisational learning, on the supply of experienced staff and on the firm's capacity to absorb the technology at all.

The contributions of the paper are as follows.

- (1) It reframes the effect of generative AI on youth employment as a question of loop dominance rather than of technology. A closed-form net entry-port elasticity decomposes the marginal effect of AI capability into displacement, reinstatement and headcount-saving components and makes loop dominance directly observable. Over the resulting design space the 25-year effect on entry employment ranges from –70.0% to +19.2% and turns positive in 17.1% of configurations, so the sign is set by two organisational design parameters, not by the model weights.
- (2) It identifies and quantifies a capability trap in workforce entry. Thinning the entry port depletes both the learning stock and the promotion flow, and because the sector's external market for experienced staff is itself fed by that promotion flow, the shortfall cannot be bought back. An automation-first strategy delivers the fastest unit-cost reduction of any scenario over the first three years yet ends the horizon with output 15.7% below the counterfactual in which the technology was never adopted — and the damage is invisible for the first five years.
- (3) It provides a direct numerical test of joint optimisation, the founding claim of socio-technical design. In a pre-specified 2×2 factorial, work redesign (technical subsystem) and investment in human–AI collaboration routines (social subsystem) interact positively by 2.74 percentage points, while a four-lever decomposition is close to additive because instruments acting on the same bottleneck substitute for one another. Complementarity in socio-technical design is a property of subsystems, not of instruments.
- (4) It is fully transparent and reproducible. The model's diffusion and augmentation parameters are calibrated to four independent published benchmarks, and the model passes the standard battery of structure, equilibrium, extreme-condition, integration-error and behaviour-reproduction tests, with a Latin hypercube sensitivity analysis over fifteen parameters. Every number in every table and figure is produced by the released code.

We believe the manuscript speaks directly to the aims of the Special Issue: it treats generative AI as a socio-technical system rather than a tool, models the feedback loops, delays and emergent consequences that connect workforce development to organisational learning and economic capability, and turns the resulting insight into actionable design and policy levers. We deeply appreciate your consideration of our manuscript and look forward to receiving comments from the reviewers. Correspondence should be directed to the address below.

Thanks very much for your attention to our paper.

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Very sincerely yours,

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